1. Write a Java program to display your name, age, and college name using variables of ;appropriate data types.

```
 public class PersonalDetails {
    public static void main(String[] args) {

        String name = "Sumit Kamal";
        int age = 20;
        String collegeName = "COER University";

        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("College Name: " + collegeName);
    }
}
```

Output:
Name: Sumit Kamal
Age: 20
College Name: COER University

2. Input two integers and print the result of all arithmetic operations (addition, subtraction, multiplication, division, and modulus).

```
 import java.util.Scanner;

public class ArithmeticOperations {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int a = sc.nextInt();

        System.out.print("Enter second integer: ");
        int b = sc.nextInt();

        System.out.println("Addition: " + (a + b));
        System.out.println("Subtraction: " + (a - b));
        System.out.println("Multiplication: " + (a * b));
        System.out.println("Division: " + (a / b));
        System.out.println("Modulus: " + (a % b));

        sc.close();
```

```
    }
}
```

Output:
Enter first integer: 10
Enter second integer: 3
Addition: 13
Subtraction: 7
Multiplication: 30
Division: 3
Modulus: 1

3. Write a Java program to check whether a given number is even or odd using the modulus operator and if-else.

```
import java.util.Scanner;

public class EvenOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        if (number % 2 == 0) {
            System.out.println(number + " is Even");
        } else {
            System.out.println(number + " is Odd");
        }

        sc.close();
    }
}
```

Output:
Enter a number: 7
7 is Odd

4:Write a program to demonstrate the use of relational operators by comparing two numbers.

```
import java.util.Scanner;
```

```
public class RelationalOperators {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.println("a == b: " + (a == b));
        System.out.println("a != b: " + (a != b));
        System.out.println("a > b: " + (a > b));
        System.out.println("a < b: " + (a < b));
        System.out.println("a >= b: " + (a >= b));
        System.out.println("a <= b: " + (a <= b));

        sc.close();
    }
}
```

Output:
Enter first number: 10
Enter second number: 20
a == b: false
a != b: true
a > b: false
a < b: true
a >= b: false
a <= b: true

5. Accept a floating-point number and convert it explicitly to an integer. Print both the original and the converted values.

```
import java.util.Scanner;

public class TypeConversion {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a floating-point number: ");
        float number = sc.nextFloat();

        // Explicit type conversion (casting)
```

```
        int convertedNumber = (int) number;

        System.out.println("Original floating-point value: " + number);
        System.out.println("Converted integer value: " + convertedNumber);

        sc.close();
    }
}
```

Output:
Enter a floating-point number: 12.75
Original floating-point value: 12.75
Converted integer value: 12

6. Write a program to accept a character and print its ASCII value.

```
import java.util.Scanner;

public class ASCIIValue {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        int ascii = (int) ch;

        System.out.println("Character: " + ch);
        System.out.println("ASCII Value: " + ascii);

        sc.close();
    }
}
```

Output:
Enter a character: A
Character: A
ASCII Value: 65

7. Input a number and check if it is divisible by both 3 and 5 using logical operators.

```
import java.util.Scanner;
```

```
public class DivisibleByThreeAndFive {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        if (number % 3 == 0 && number % 5 == 0) {
            System.out.println(number + " is divisible by both 3 and 5.");
        } else {
            System.out.println(number + " is not divisible by both 3 and 5.");
        }

        sc.close();
    }
}
```

Output:
Enter a number: 15
15 is divisible by both 3 and 5.

8. Input a character and check whether it is a vowel or a consonant using if-else.

```
import java.util.Scanner;

public class VowelConsonant {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {

            System.out.println(ch + " is a Vowel");
        } else {
            System.out.println(ch + " is a Consonant");
        }

        sc.close();
    }
}
```

Output:
Enter a character: A
A is a Vowel

9. Input marks of three subjects, calculate total and percentage, and print pass/fail using if-else (passing criteria: >=40% in all subjects).

```
import java.util.Scanner;

public class MarksResult {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter marks of Subject 1: ");
        double sub1 = sc.nextDouble();

        System.out.print("Enter marks of Subject 2: ");
        double sub2 = sc.nextDouble();

        System.out.print("Enter marks of Subject 3: ");
        double sub3 = sc.nextDouble();

        double total = sub1 + sub2 + sub3;
        double percentage = total / 3;

        System.out.println("Total Marks: " + total);
        System.out.println("Percentage: " + percentage + "%");

        if (sub1 >= 40 && sub2 >= 40 && sub3 >= 40) {
            System.out.println("Result: Pass");
        } else {
            System.out.println("Result: Fail");
        }

        sc.close();
    }
}
```

Output:
Enter marks of Subject 1: 70
Enter marks of Subject 2: 65
Enter marks of Subject 3: 80
Total Marks: 215.0
Percentage: 71.66666666666667%

Result: Pass

10. Write a program to find the greatest of two numbers using if-else.

```
import java.util.Scanner;

public class GreatestOfTwo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int num1 = sc.nextInt();

        System.out.print("Enter second number: ");
        int num2 = sc.nextInt();

        if (num1 > num2) {
            System.out.println(num1 + " is the greatest number.");
        } else if (num2 > num1) {
            System.out.println(num2 + " is the greatest number.");
        } else {
            System.out.println("Both numbers are equal.");
        }

        sc.close();
    }
}
```

Output:
Enter first number: 25
Enter second number: 18
25 is the greatest number.

Intermediate Level (11–20)

11. Input three numbers and print the largest using nested if-else.

```
 import java.util.Scanner;

public class LargestOfThree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter first number: ");
        int num1 = sc.nextInt();

        System.out.print("Enter second number: ");
        int num2 = sc.nextInt();

        System.out.print("Enter third number: ");
        int num3 = sc.nextInt();

        if (num1 >= num2) {
            if (num1 >= num3) {
                System.out.println(num1 + " is the largest number.");
            } else {
                System.out.println(num3 + " is the largest number.");
            }
        } else {
            if (num2 >= num3) {
                System.out.println(num2 + " is the largest number.");
            } else {
                System.out.println(num3 + " is the largest number.");
            }
        }

        sc.close();
    }
}
```

Output:
Enter first number: 10
Enter second number: 25
Enter third number: 18
25 is the largest number.

12. Accept a year and check whether it is a leap year using if-else. import java.util.Scanner;

```
public class LeapYear {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a year: ");
        int year = sc.nextInt();
```

```
        if (year % 400 == 0) {
            System.out.println(year + " is a Leap Year.");
        } else if (year % 100 == 0) {
            System.out.println(year + " is not a Leap Year.");
        } else if (year % 4 == 0) {
            System.out.println(year + " is a Leap Year.");
        } else {
            System.out.println(year + " is not a Leap Year.");
        }

        sc.close();
    }
}
```

Output:
Enter a year: 2024
2024 is a Leap Year.

13. Accept a number and check whether it is positive, negative, or zero using if-else.

```
import java.util.Scanner;

public class PositiveNegativeZero {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        if (number > 0) {
            System.out.println(number + " is Positive.");
        } else if (number < 0) {
            System.out.println(number + " is Negative.");
        } else {
            System.out.println("The number is Zero.");
        }

        sc.close();
    }
}
```

Output:
Enter a number: -15

-15 is Negative.

14. Accept a character and check if it is a digit, uppercase letter, lowercase letter, or a special character using character classification and if-else.

```
import java.util.Scanner;

public class CharacterClassification {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        if (ch >= '0' && ch <= '9') {
            System.out.println(ch + " is a Digit.");
        }
        else if (ch >= 'A' && ch <= 'Z') {
            System.out.println(ch + " is an Uppercase Letter.");
        }
        else if (ch >= 'a' && ch <= 'z') {
            System.out.println(ch + " is a Lowercase Letter.");
        }
        else {
            System.out.println(ch + " is a Special Character.");
        }

        sc.close();
    }
}
```

Output:
Enter a character: @
@ is a Special Character.

15. Accept a 3-digit number and check whether it is an Armstrong number using arithmetic operators and if-else.

```
import java.util.Scanner;

public class ArmstrongNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a 3-digit number: ");
        int number = sc.nextInt();

        int originalNumber = number;

        int digit1 = number % 10;
        number = number / 10;

        int digit2 = number % 10;
        number = number / 10;

        int digit3 = number % 10;

        int sum = (digit1 * digit1 * digit1)
                + (digit2 * digit2 * digit2)
                + (digit3 * digit3 * digit3);

        if (sum == originalNumber) {
            System.out.println(originalNumber + " is an Armstrong Number.");
        } else {
            System.out.println(originalNumber + " is not an Armstrong Number.");
        }

        sc.close();
    }
}
```

Output:
Enter a 3-digit number: 153
153 is an Armstrong Number.

16. Input salary and calculate tax based on income slabs using if-else:
Income ≤ 250000: No tax
250001 to 500000: 5%
500001 to 1000000: 20%
Above 1000000: 30%

```
import java.util.Scanner;

public class IncomeTax {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter your annual salary: ");
        double salary = sc.nextDouble();

        double tax;

        if (salary <= 250000) {
            tax = 0;
        } else if (salary <= 500000) {
            tax = salary * 0.05;
        } else if (salary <= 1000000) {
            tax = salary * 0.20;
        } else {
            tax = salary * 0.30;
        }

        System.out.println("Annual Salary: " + salary);
        System.out.println("Income Tax: " + tax);

        sc.close();
    }
}
```

Output:
Enter your annual salary: 600000
Annual Salary: 600000.0
Income Tax: 120000.0

17. Accept an integer and check whether it is a palindrome using arithmetic operations and type casting.

```
import java.util.Scanner;

public class PalindromeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int number = sc.nextInt();

        int originalNumber = number;
        int reverse = 0;

        while (number != 0) {
            int digit = number % 10;
```

```
            reverse = reverse * 10 + digit;
            number = number / 10;
        }

        if (originalNumber == reverse) {
            System.out.println(originalNumber + " is a Palindrome Number.");
        } else {
            System.out.println(originalNumber + " is not a Palindrome Number.");
        }

        sc.close();
    }
}
```

Output:
Enter an integer: 121
121 is a Palindrome Number.

18. Accept a number and print whether it is prime or not using if-else.

```
import java.util.Scanner;

public class PrimeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        boolean isPrime = true;

        if (number <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i <= number / 2; i++) {
                if (number % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        if (isPrime) {
```

```
            System.out.println(number + " is a Prime Number.");
        } else {
            System.out.println(number + " is not a Prime Number.");
        }

        sc.close();
    }
}
```

Output:
Enter a number: 17
17 is a Prime Number.

19. Accept a character and use if-else to identify whether it's a valid operator (+, -, *, /) and perform the corresponding operation on two input numbers.

```
import java.util.Scanner;

public class CalculatorOperator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();

        System.out.print("Enter an operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        if (operator == '+') {
            System.out.println("Result = " + (num1 + num2));
        } else if (operator == '-') {
            System.out.println("Result = " + (num1 - num2));
        } else if (operator == '*') {
            System.out.println("Result = " + (num1 * num2));
        } else if (operator == '/') {
            if (num2 != 0) {
                System.out.println("Result = " + (num1 / num2));
            } else {
                System.out.println("Division by zero is not allowed.");
            }
```

```
        } else {
            System.out.println("Invalid Operator!");
        }

        sc.close();
    }
}
```

Output:
Enter first number: 10
Enter second number: 5
Enter an operator (+, -, *, /): *
Result = 50.0

20. Accept age and gender and determine eligibility for marriage (male ≥ 21, female ≥ 18) using nested if-else conditions.

```
import java.util.Scanner;

public class MarriageEligibility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        System.out.print("Enter your gender (M/F): ");
        char gender = sc.next().charAt(0);

        if (gender == 'M' || gender == 'm') {
            if (age >= 21) {
                System.out.println("Eligible for marriage.");
            } else {
                System.out.println("Not eligible for marriage.");
            }
        } else if (gender == 'F' || gender == 'f') {
            if (age >= 18) {
                System.out.println("Eligible for marriage.");
            } else {
                System.out.println("Not eligible for marriage.");
            }
        } else {
            System.out.println("Invalid gender entered.");
```

```
        }

        sc.close();
    }
}
```

Output:
Enter your age: 22
Enter your gender (M/F): M
Eligible for marriage.

Advanced Level (21–30)

21. Accept three angles and check whether they can form a valid triangle (sum of angles = 180°).

```
import java.util.Scanner;

public class ValidTriangle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first angle: ");
        int angle1 = sc.nextInt();

        System.out.print("Enter second angle: ");
        int angle2 = sc.nextInt();

        System.out.print("Enter third angle: ");
        int angle3 = sc.nextInt();

        if (angle1 + angle2 + angle3 == 180) {
            System.out.println("The angles can form a valid triangle.");
        } else {
            System.out.println("The angles cannot form a valid triangle.");
        }

        sc.close();
    }
}
```

Output:
Enter first angle: 60

Enter second angle: 70
Enter third angle: 50
The angles can form a valid triangle.

22. Input the length and breadth of a rectangle, and determine whether it is a square using if-else.

```
import java.util.Scanner;

public class SquareCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter length: ");
        double length = sc.nextDouble();

        System.out.print("Enter breadth: ");
        double breadth = sc.nextDouble();

        if (length == breadth) {
            System.out.println("The rectangle is a Square.");
        } else {
            System.out.println("The rectangle is not a Square.");
        }

        sc.close();
    }
}
```

Output:
Enter length: 10
Enter breadth: 10
The rectangle is a Square.

23. Input a temperature in Celsius and convert it to Fahrenheit using type casting and expressions.

```
import java.util.Scanner;

public class CelsiusToFahrenheit {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter temperature in Celsius: ");
        double celsius = sc.nextDouble();

        double fahrenheit = (celsius * 9 / 5) + 32;

        System.out.println("Temperature in Celsius: " + celsius);
        System.out.println("Temperature in Fahrenheit: " + fahrenheit);

        sc.close();
    }
}
```

Output:
Enter temperature in Celsius: 25
Temperature in Celsius: 25.0
Temperature in Fahrenheit: 77.0

24. Accept a number and check whether it is a perfect number (sum of divisors = number) using basic arithmetic and if-else.

```
import java.util.Scanner;

public class PerfectNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        int sum = 0;

        for (int i = 1; i < number; i++) {
            if (number % i == 0) {
                sum = sum + i;
            }
        }

        if (sum == number && number > 0) {
            System.out.println(number + " is a Perfect Number.");
        } else {
            System.out.println(number + " is not a Perfect Number.");
        }
```

```
        sc.close();
    }
}
```

Output:
Enter a number: 28
28 is a Perfect Number.

25. Accept two integers and a logical operator as a character. Use if-else to perform the logical operation and display the result.

```
import java.util.Scanner;

public class LogicalOperations {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int num1 = sc.nextInt();

        System.out.print("Enter second integer: ");
        int num2 = sc.nextInt();

        System.out.print("Enter logical operator (&, |): ");
        char operator = sc.next().charAt(0);

        boolean result;

        if (operator == '&') {
            result = (num1 != 0 && num2 != 0);
            System.out.println("Result: " + result);
        } else if (operator == '|') {
            result = (num1 != 0 || num2 != 0);
            System.out.println("Result: " + result);
        } else {
            System.out.println("Invalid logical operator.");
        }

        sc.close();
    }
}
```

Output:
Enter first integer: 1

Enter second integer: 0
Enter logical operator (&, |): &
Result: false

26. Input a five-digit number and check whether the first and last digits are the same using arithmetic expressions.

```
import java.util.Scanner;

public class FirstLastDigit {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a five-digit number: ");
        int number = sc.nextInt();

        int lastDigit = number % 10;
        int firstDigit = number / 10000;

        if (firstDigit == lastDigit) {
            System.out.println("First and last digits are the same.");
        } else {
            System.out.println("First and last digits are different.");
        }

        sc.close();
    }
}
```

Output:
Enter a five-digit number: 12321
First and last digits are the same.

27. Input the cost price and selling price of an item. Determine profit or loss, and by how much, using if-else.

```
import java.util.Scanner;

public class ProfitLoss {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Cost Price: ");
        double costPrice = sc.nextDouble();

        System.out.print("Enter Selling Price: ");
        double sellingPrice = sc.nextDouble();

        if (sellingPrice > costPrice) {
            double profit = sellingPrice - costPrice;
            System.out.println("Profit: " + profit);
        } else if (costPrice > sellingPrice) {
            double loss = costPrice - sellingPrice;
            System.out.println("Loss: " + loss);
        } else {
            System.out.println("No Profit, No Loss.");
        }

        sc.close();
    }
}
```

Output:
Enter Cost Price: 500
Enter Selling Price: 650
Profit: 150.0

28. Accept a number and display the number of digits using expressions and if-else.

```
import java.util.Scanner;

public class NumberOfDigits {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        number = Math.abs(number);

        if (number < 10) {
            System.out.println("Number of digits: 1");
        } else if (number < 100) {
            System.out.println("Number of digits: 2");
```

```
        } else if (number < 1000) {
            System.out.println("Number of digits: 3");
        } else if (number < 10000) {
            System.out.println("Number of digits: 4");
        } else if (number < 100000) {
            System.out.println("Number of digits: 5");
        } else {
            System.out.println("Number has 6 or more digits.");
        }

        sc.close();
    }
}
```

Output:
Enter a number: 12345
Number of digits: 5

29. Input the height (in cm) of a person and classify them as short (<150), average (150–170), or tall (>170).

```
import java.util.Scanner;

public class HeightClassification {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter height in cm: ");
        double height = sc.nextDouble();

        if (height < 150) {
            System.out.println("Person is Short.");
        } else if (height <= 170) {
            System.out.println("Person is Average.");
        } else {
            System.out.println("Person is Tall.");
        }

        sc.close();
    }
}
```

Output:
Enter height in cm: 175
Person is Tall.

30. Write a Java program to simulate a simple calculator that takes two numbers and an operator (+, , *, /) and performs the corresponding operation using if-else.

```
import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        double num1 = sc.nextDouble();

        System.out.print("Enter second number: ");
        double num2 = sc.nextDouble();

        System.out.print("Enter an operator (+, -, *, /): ");
        char operator = sc.next().charAt(0);

        if (operator == '+') {
            System.out.println("Result = " + (num1 + num2));
        }
        else if (operator == '-') {
            System.out.println("Result = " + (num1 - num2));
        }
        else if (operator == '*') {
            System.out.println("Result = " + (num1 * num2));
        }
        else if (operator == '/') {
            if (num2 != 0) {
                System.out.println("Result = " + (num1 / num2));
            } else {
                System.out.println("Error: Division by zero is not allowed.");
            }
        }
        else {
            System.out.println("Invalid operator!");
        }

        sc.close();
```

```
    }
}
```

Output:
Enter first number: 20
Enter second number: 4
Enter an operator (+, -, *, /): /
Result = 5.0